AI-Driven Probabilistic Models for Robust Risk-Based Household Battery Scheduling

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Household battery scheduling: why probabilistic energy price forecasting?

The Context: Home batteries need optimal scheduling decisions

- ✓ When to charge? When to discharge?
- ✓ Profitability depends on accurate day-ahead price forecasts

The Challenge: Increased price volatility

- ✓ Renewable energy integration → unpredictable price spikes
- ✓ Traditional point forecasts fail to capture uncertainty
- ✓ Critical information lost in extreme price events

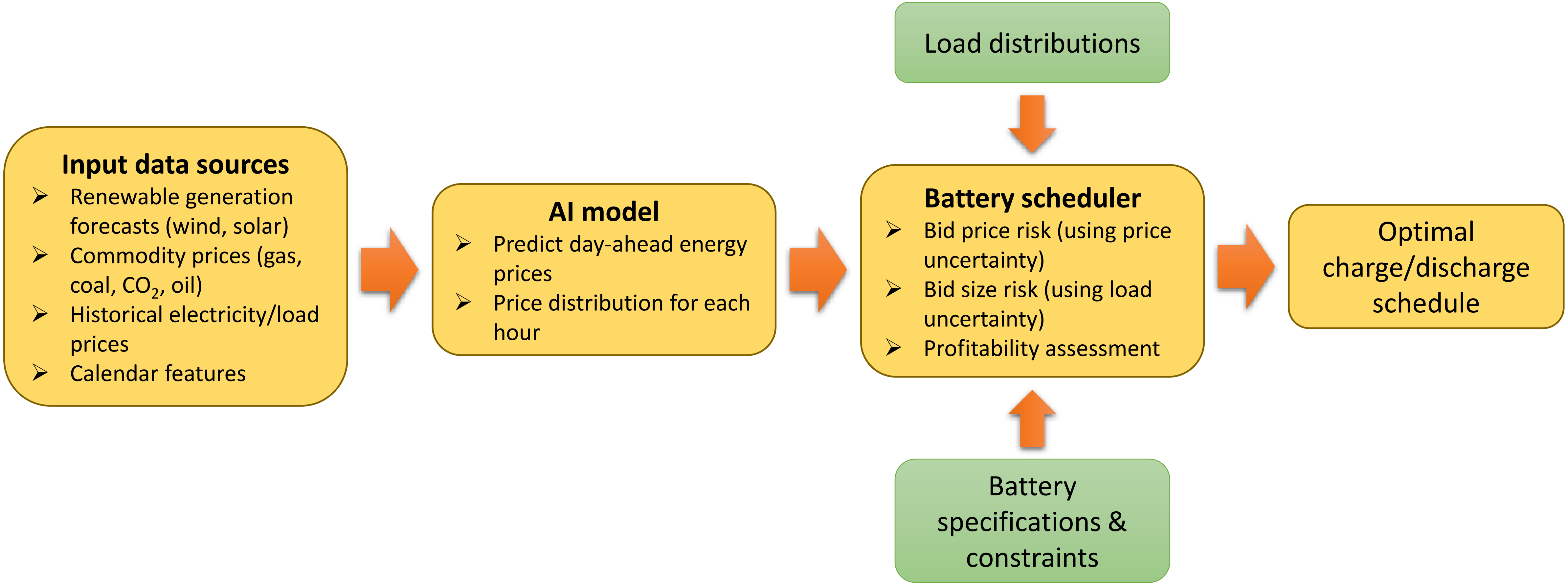
The Gap: Need for probabilistic price forecasts

- ✓ Capture full distribution of possible prices
- ✓ Quantify uncertainty for risk-aware decision making
- ✓ Enable robust battery scheduling strategies

The solution: risk-based scheduling using probabilistic forecasts

- ✓ Multi-horizon forecasting (day-ahead, 48 hours)
- ✓ Evaluation on Dutch EPEX market

End-to-end AI-based probabilistic forecasting pipeline for battery scheduling



Problem Formulation

INPUTS: Feature Representation at time t

Features from relevant time indices:

$$\mathbf{X}_t = \{\mathbf{x}_\tau : \tau \in \mathcal{T}_t\}$$

where $\mathcal{T}_t$ = set of time indices (e.g., $\{t, t-1, t-2, t-7\}$) and each $\mathbf{x}_\tau \in \mathbb{R}^F$ contains:

- Historical prices: p_τ
- Renewable generation forecasts: solar, wind (onshore/offshore)
- Commodity prices: gas, coal, CO_2
- Load forecast, calendar features

Note: Lag structure $\mathcal{T}_t$ is model-dependent

TARGET: Multi-Horizon Price Forecasting

For forecast horizon $h \in \{1, 2, \ldots, H\}$ where $H = 24$ hours

Predict: $\mathbb{P}(p_{t+h} \mid \mathbf{X}_t)$

where:

- $p_{t+h} \in \mathbb{R}$ = electricity price at time $t+h$ [EUR/MWh]
- $\mathbf{X}_t$ = feature representation (from relevant historical time points)

PROBABILISTIC OUTPUT

Each model $f : \mathbf{X}_t \to$ distributional forecast

(a) Quantile forecasts:

$$\hat{q}_\alpha(p_{t+h} \mid \mathbf{X}_t) \text{ for } \alpha \in \{0.05, 0.25, 0.50, 0.75, 0.95\}$$

where $\mathbb{P}(p_{t+h} \le \hat{q}_\alpha) = \alpha$

(b) Parametric: NGBoost (Normal distribution)

All models: $\mathbf{X}_t \mapsto \hat{\mathbb{P}}(p_{t+h} \mid \mathbf{X}_t)$ for $h = 1, \ldots, 24$

Note: Each method may use different subsets of $\mathbf{X}_t$ and different lag structures $\mathcal{T}_t$

Naive

CONCEPT: Persistence Forecast with Weekly Seasonality

Idea: Electricity prices exhibit strong weekly patterns. Use same-hour price from last week.

Point Forecast:

$$\hat{y}_{t+h} = y_{t+h-168}, \qquad h = 1, 2, \ldots, 24$$

where 168 hours = 7 days (weekly lag)

Implementation:

- Weekend adjustment: Friday same-hour for Sat/Sun
- No training required (rule-based)
- Baseline benchmark for comparison

PROBABILISTIC OUTPUT

Quantile Forecasts:

$$\hat{q}_\tau(t) = \hat{y}_t + Q_\tau(\mathcal{E}_{\text{hist}})$$

where $Q_\tau(\mathcal{E}_{\text{hist}})$ = τ-th quantile of historical forecast errors
Each $\tau \in \{0.01, 0.05, 0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9, 0.95, 0.99\}$ produces independent offset

Quantile Regression Averaging (QRA)

CONCEPT: Direct Conditional Quantile Estimation

Linear quantile regression without regularization

For each $\tau \in \{0.01, 0.05, \ldots, 0.99\}$, solve:

$$\hat{\beta}_\tau = \arg\min_\beta \sum_{i=1}^{n} \rho_\tau(y_i - \mathbf{x}_i^T \beta)$$

Pinball Loss:

$$\rho_\tau(u) = \begin{cases} \tau \cdot u & \text{if } u \geq 0 \\ (\tau - 1) \cdot u & \text{if } u < 0 \end{cases}$$

Key Properties:

- Solver: Linear programming (HiGHS)
- No regularization ($\lambda = 0$)
- Independent model per quantile

PROBABILISTIC OUTPUT

$\hat{q}_\tau(t) = \mathbf{x}_t^T \hat{\beta}_\tau$ for all τ produces **13 quantile forecasts** forming complete predictive distribution

Forecasting Models

Lasso QRA

CONCEPT: L1-Regularized Quantile Regression with Feature Selection

For each τ, solve regularized optimization:

$$\hat{\beta}_\tau(\lambda) = \arg\min_{\beta} \left[\sum_{i=1}^{n} \rho_\tau(y_i - \mathbf{x}_i^T \beta) + \lambda \|\beta\|_1 \right]$$

where $\|\beta\|_1 = \sum_{j=1}^{p} |\beta_j|$ (L1/Lasso penalty)

BIC Selection (per quantile):

$$\text{BIC}(\lambda, \tau) = \log\left(\frac{1}{n} \sum_{i=1}^{n} \rho_\tau(y_i - \hat{y}_i) \right) + \frac{m(\lambda) \cdot \log(n)}{2n \cdot \log(p)}$$

Select: $\hat{\lambda}_\tau^* = \arg\min_\lambda \text{BIC}(\lambda, \tau)$ where $m(\lambda) = \#$ non-zero coefficients

Implementation:

- Lambda grid: 30 values, $\{\lambda_i = 10^{2(i-1)/9}\}_{i=1}^{29} \cup \{0\}$ (range: 0 to 100)
- BIC computed on training data (no CV)
- Independent λ^* per τ

KEY ADVANTAGE

Automatic feature selection yields sparse, interpretable models while preventing overfitting

Natural Gradient Boosting (NGBoost)

CONCEPT: Probabilistic Boosting via Natural Gradients

Predicts full Gaussian distribution $P(y \mid \mathbf{x}) = \mathcal{N}(\mu(\mathbf{x}), \sigma^2(\mathbf{x}))$

Boosting Updates: Learn parameters $\boldsymbol{\theta} = (\mu, \sigma)$ iteratively:

$$\boldsymbol{\theta}_m = \boldsymbol{\theta}_{m-1} + \eta \cdot f_m(\mathbf{x}), \quad m = 1, 2, \ldots, M$$

where f_m = decision tree fitted to natural gradient: $\tilde{\nabla}_{\boldsymbol{\theta}} \mathcal{L} = \tilde{\nabla}_{\boldsymbol{\theta}}[-\log P(y \mid \boldsymbol{\theta}(\mathbf{x}))]$

Hyperparameters:

- $\#$ estimators
- Learning rate: η
- Tree depth
- Base learner: Decision tree (Friedman MSE)

PROBABILISTIC OUTPUT

Point: $\hat{y}(t) = \mu_M(\mathbf{x}_t)$ | **Quantile:** $\hat{q}_\tau(t) = \mu_M(\mathbf{x}_t) + \sigma_M(\mathbf{x}_t) \cdot \Phi^{-1}(\tau)$
Full Gaussian distribution where Φ^{-1} = inverse standard normal CDF

EPF Pipeline: AI Workflow

Data acquisition & feature engineering → Feature selection → **AI model tuning & training** → Day-ahead energy price prediction

EPF Pipeline: AI Workflow

Data acquisition
&
feature engineering

EPF Pipeline: AI Workflow

Dutch EPEX market

Dataset acquisition

- ✓ Time frame: hourly data from 2016-01-01 00:00:00 to 2024-08-31 23:00:00 (~76,000 hours)
- ✓ Frequency: 1-hour intervals, continuous 24/7 series

Features

Day index (seasonal pattern)
Yearly cyclical encoding (sine component)
Yearly cyclical encoding (cosine component)
Hour index 0–23
Daily cyclical encoding (sine component)
Daily cyclical encoding (cosine component)
Target variable (Dutch EPEX prices) [EUR/MWh]
Predicted national electrical demand [MW]
Observed national demand [MW]
Forecasted PV output [MW]
Onshore wind forecast [MW]
Offshore wind forecast [MW]
Sum of onshore + offshore wind [MW]
Total predicted power generation [MW]
Dutch natural gas benchmark price [EUR/MWh]
EU carbon price [EUR/tCO_2]
Coal benchmark price [EUR/t]
Oil benchmark price [EUR/bbl]

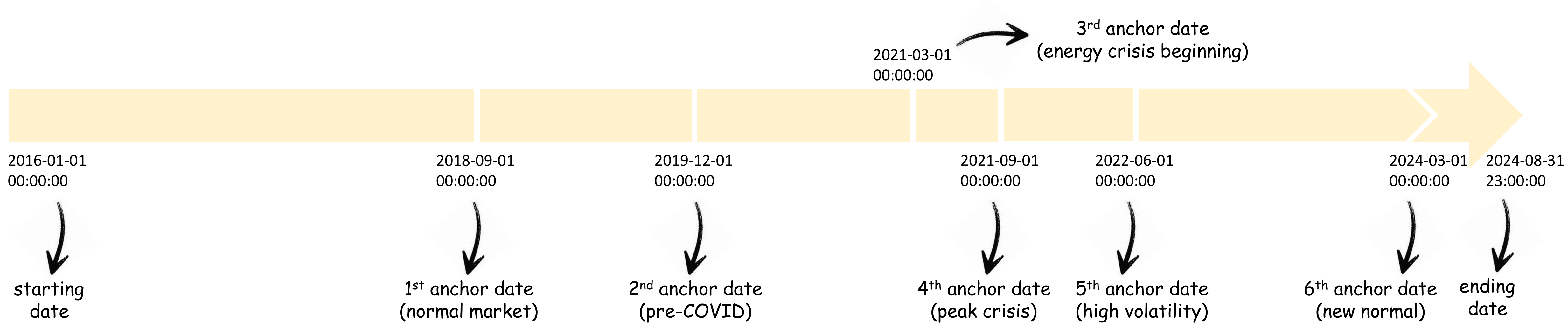
EPF Pipeline: AI Workflow

Feature selection → AI model tuning & training

EPF Pipeline: AI Workflow

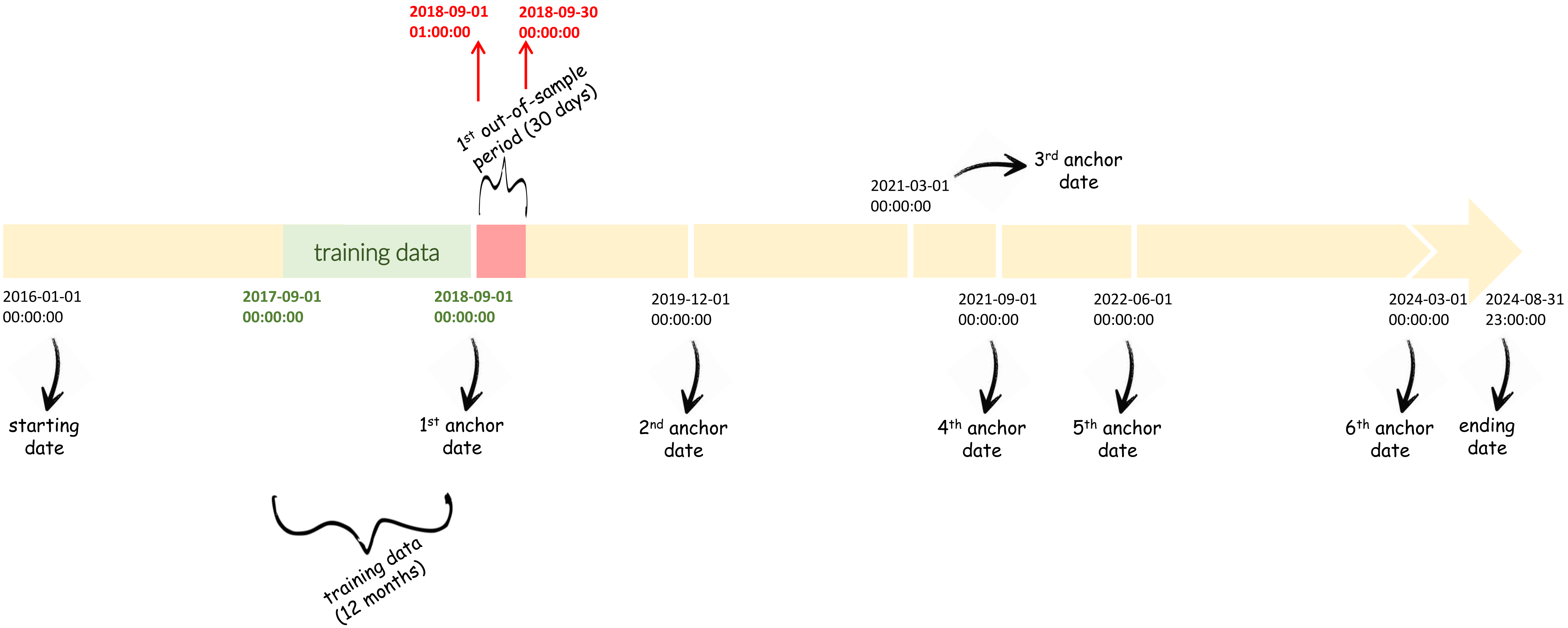
Why to use anchor dates?

- Electricity markets evolve over time, and older patterns gradually lose relevance
- Anchor dates define fixed recalibration points to keep forecasts aligned with current market regimes
- Periodic retraining keeps the models aligned with evolving market conditions
- Anchor-based design produces robust out-of-sample evaluation windows



EPF Pipeline: AI Workflow

First anchor window setup



EPF Pipeline: AI Workflow

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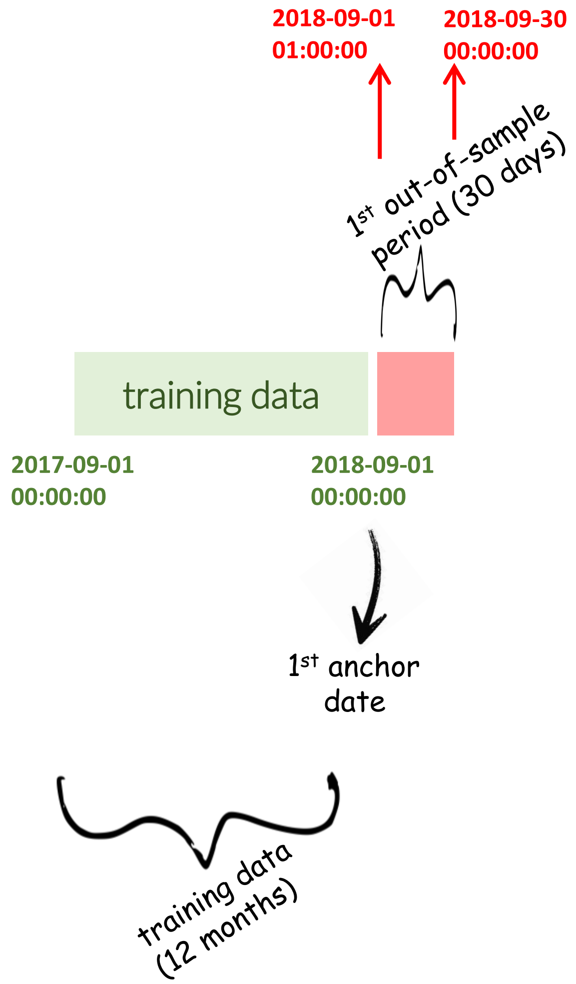
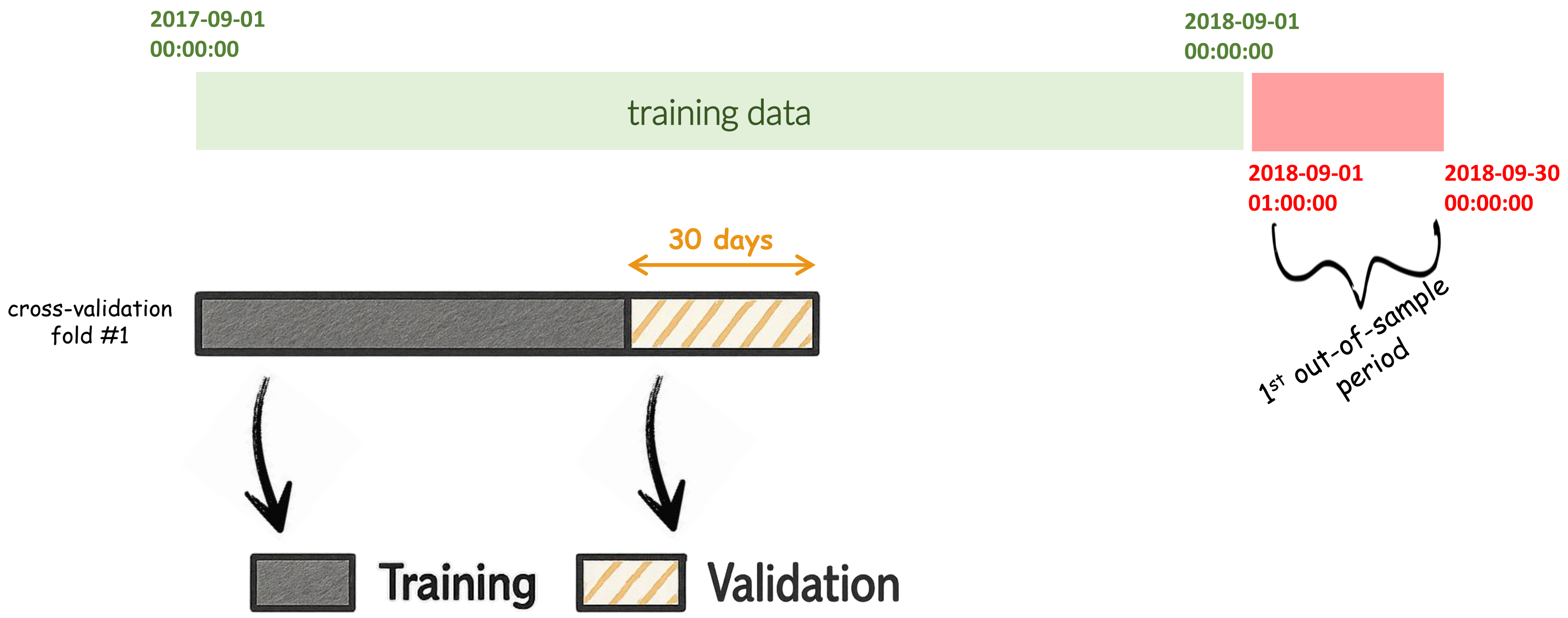


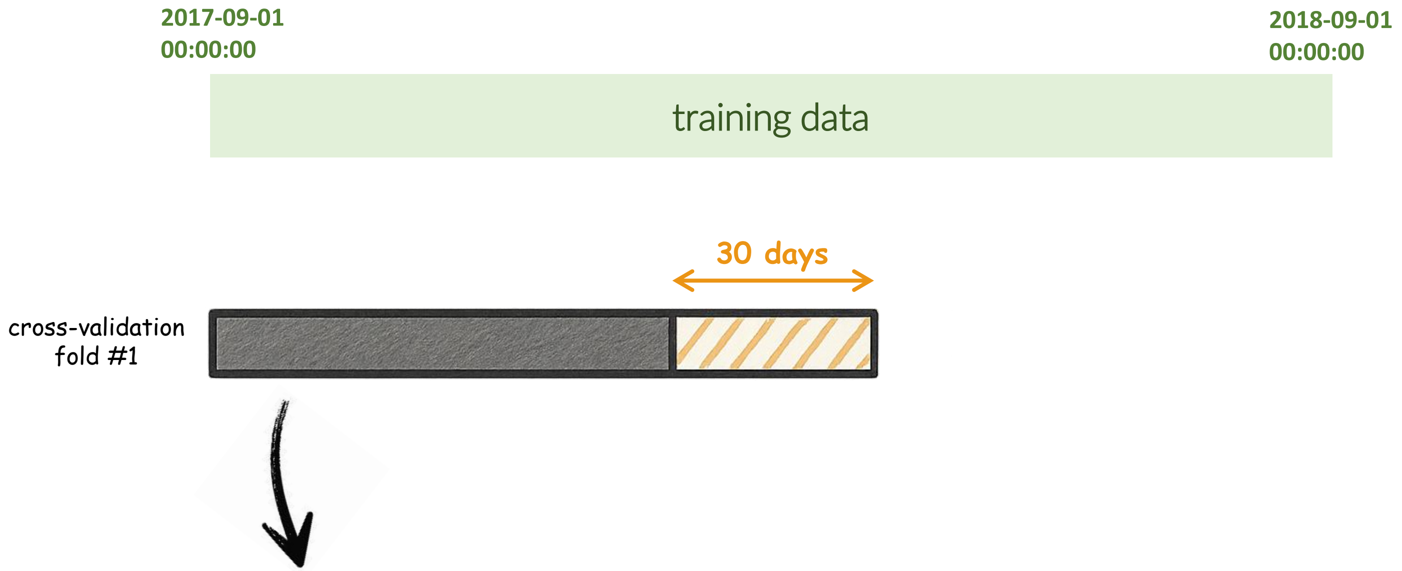
EPF Pipeline: AI Workflow

Cross-validation & feature selection



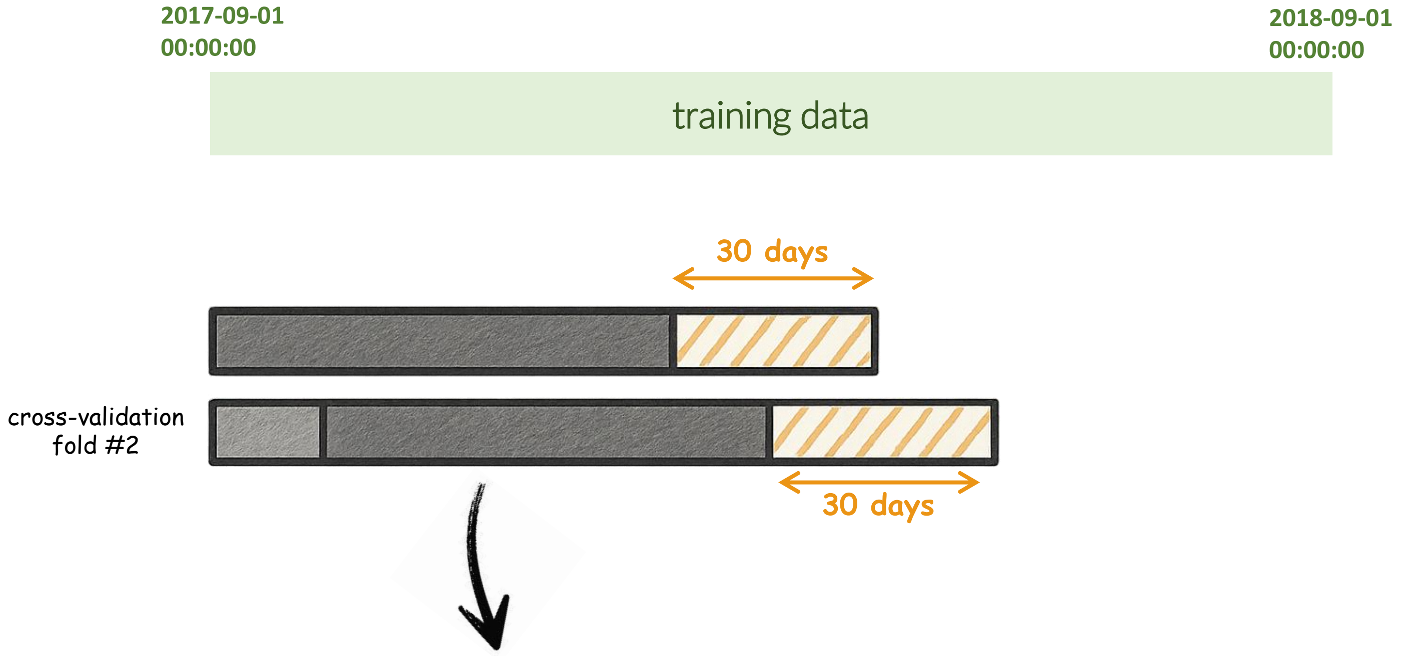
EPF Pipeline: AI Workflow

Cross-validation & feature selection



- » Normalize/scale the features
- » Compute mutual information or correlation scores, between the training features and the training target data
- » Select top-K features, e.g., K=20

EPF Pipeline: AI Workflow



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EPF Pipeline: AI Workflow

Cross-validation &
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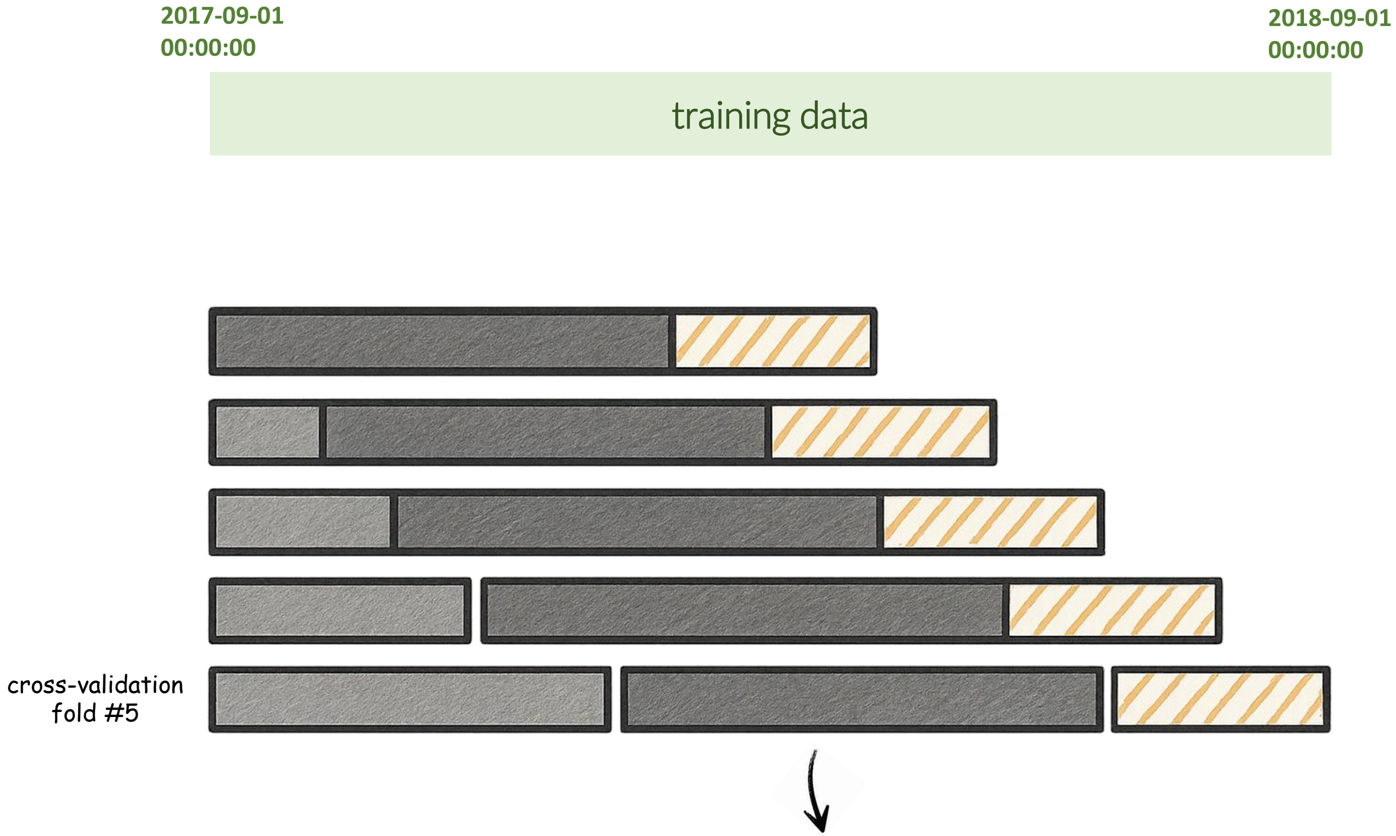

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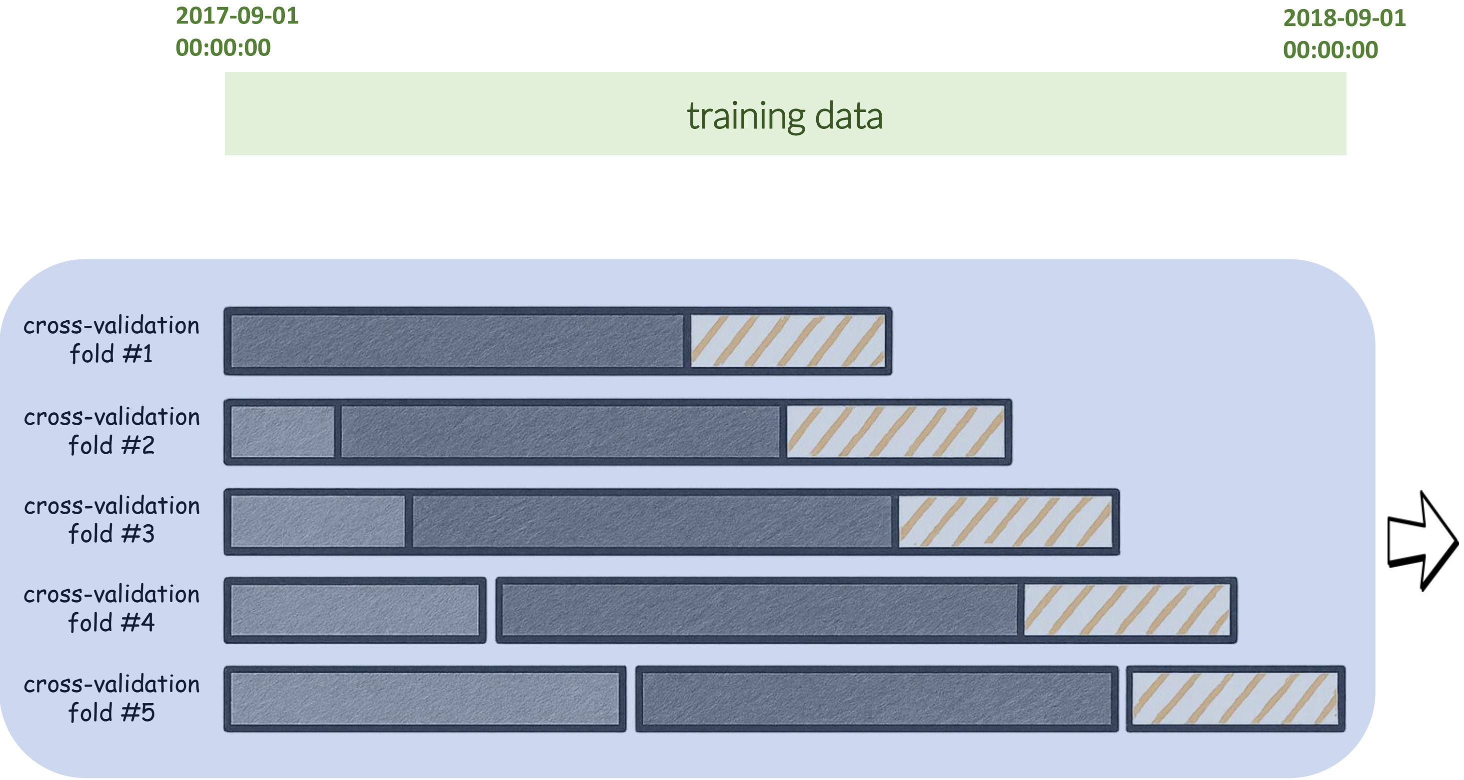
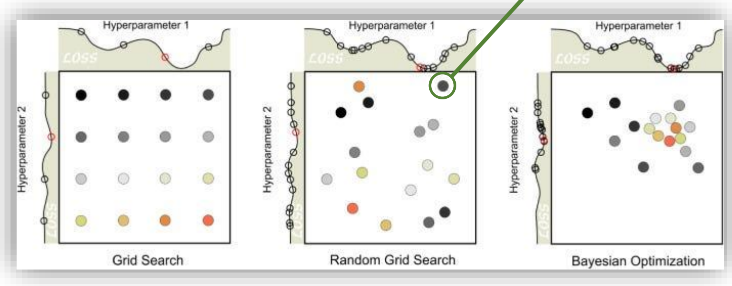
EPF Pipeline: AI Workflow



- Collect feature selections from 5 folds
- Apply majority voting: count frequency of each feature across folds
- Final feature set: select top-N (e.g. N=60) features with highest frequencies (ties broken by average mutual information if needed)

EPF Pipeline: AI Workflow



One hyper-parameter combination at a time

2017-09-01 00:00:00
2018-09-01 00:00:00

training data→ final features set

2018-09-01 01:00:00
2018-09-30 00:00:00

1st out-of-sample period

fold #1 → Performance Metric$_1$
fold #2 → Performance Metric$_2$
fold #3 → Performance Metric$_3$
fold #4 → Performance Metric$_4$
fold #5 → Performance Metric$_5$

Average

Dropped | Training | Validation

Best combination of hyper-parameters

2018-09-01 01:00:00
2018-09-30 00:00:00

1st out-of-sample period load values

training data→ final features set

20217-09-01 00:00:00
2018-09-01 00:00:00

fit()

Best model

predict()

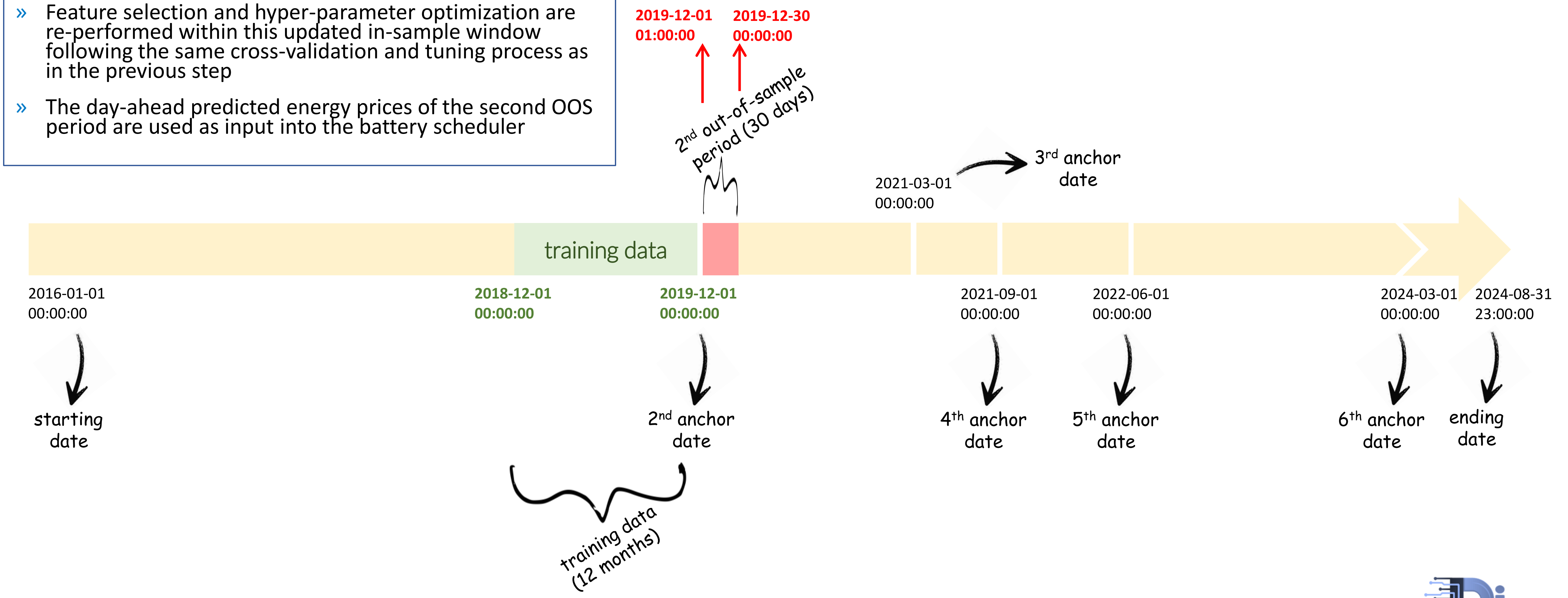
Energy prices predictions

Battery scheduler

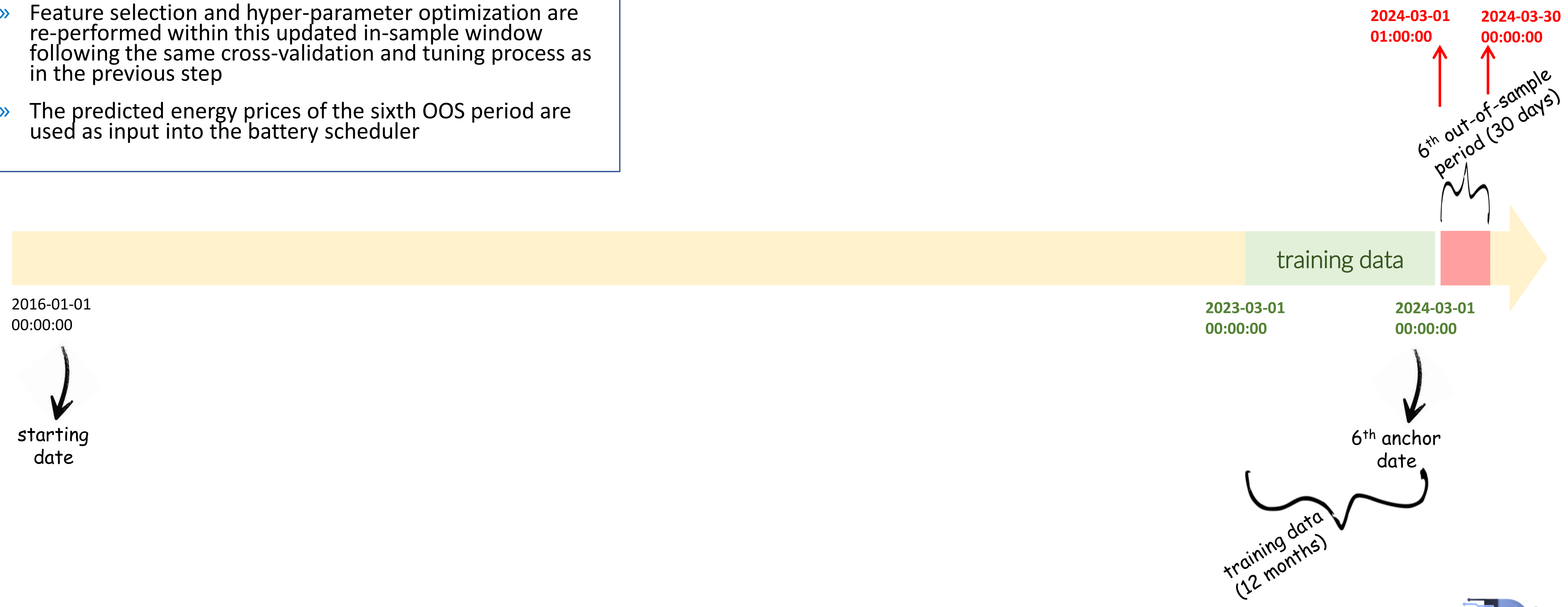
Charge/Discharge decision

EPF Pipeline: AI Workflow

- The same walk-forward procedure is applied for the second out-of-sample (OOS) period
- The training window now includes data up to the 2nd anchor date, maintaining the same historical length as before (12 months)
- Feature selection and hyper-parameter optimization are re-performed within this updated in-sample window following the same cross-validation and tuning process as in the previous step
- The day-ahead predicted energy prices of the second OOS period are used as input into the battery scheduler



- The same walk-forward procedure is applied for all the rest OOS periods until the last (sixth) OOS period
- The training window now includes data up to the 6th anchor date, maintaining the same historical length as before (12 months)
- Feature selection and hyper-parameter optimization are re-performed within this updated in-sample window following the same cross-validation and tuning process as in the previous step
- The predicted energy prices of the sixth OOS period are used as input into the battery scheduler



Experimental Results

Settings

Naive

- ✓ Quantiles: τ = {0.01, 0.05, 0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9, 0.95, 0.99}
- ✓ No hyper-parameter tuning
- ✓ Key parameters: last-week same-hour forecast

QRA

- ✓ Quantiles: τ = {0.01, 0.05, 0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9, 0.95, 0.99}
- ✓ No hyper-parameter tuning
- ✓ λ=0 (no regularization)

LQRA (BIC)

- ✓ Quantiles: τ = {0.01, 0.05, 0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9, 0.95, 0.99}
- ✓ No hyper-parameter tuning
- ✓ 30 λ values logarithmically spaced from 0.01 to 100: best λ per quantile

NGBoost

- ✓ Continuous distribution
- ✓ 3-fold rolling window cross-validation
- ✓ Number of estimators: {100,200,300}
- ✓ Learning rate: {0.01,0.05,0.1}
- ✓ Maximum depth: {3,4,5}

Feature Created

- ✓ Lag features: 1, 2, 5, 7 days
- ✓ Rolling statistics: 24h and 168h windows (mean, std, min, max)
- ✓ Daily aggregates: Previous day min/max/mean/std
- ✓ No feature selection: all engineered features used

Experimental Results

» Data preprocessing

- Target transformation: $\operatorname{asinh}(y) = \log\left(y + \sqrt{y^2 + 1}\right)$
- Feature scaling is robust standardization: $x_{scaled} = \dfrac{x - \overbrace{median(X)}^{\text{entire dataset}}}{\underbrace{MAD(X)}_{MAD(X) = median(|x_i - median(X)|)} \times \underbrace{1.4826}_{\text{scaling constant}}}$

» Evaluation metrics

- MAE (Mean Absolute Error): average magnitude of forecast errors in EUR/MWh
- RMSE (Root Mean Squared Error): square root of average squared errors

Point forecast metrics

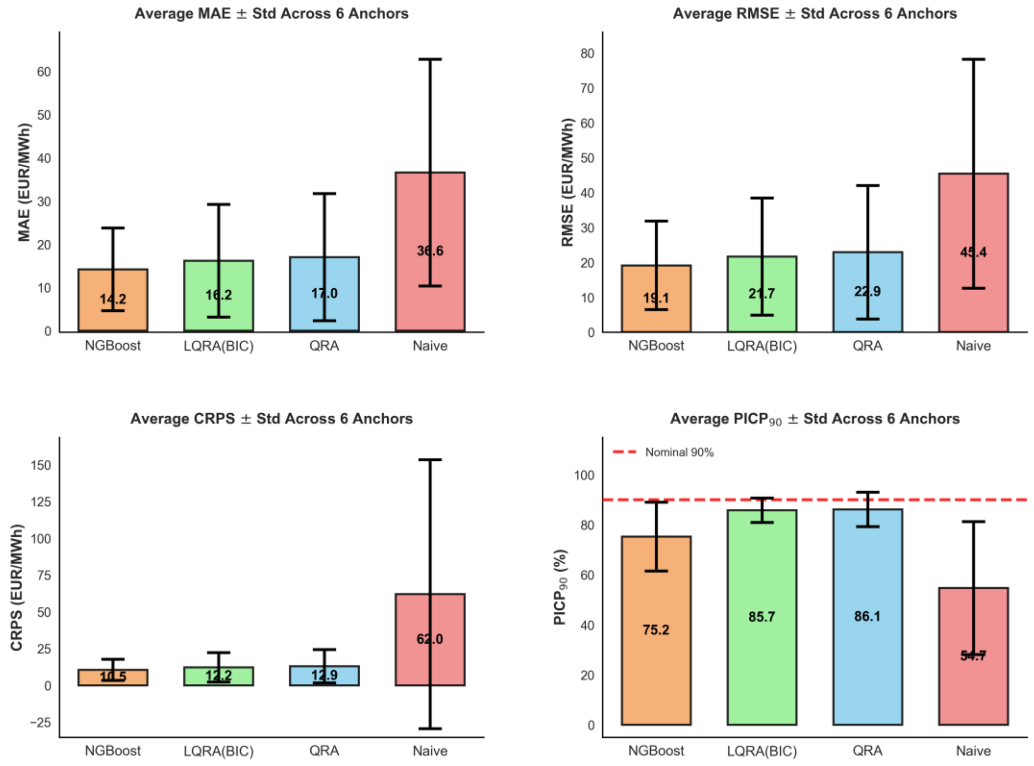
- CRPS (Continuous Ranked Probability Score, EUR/MWh)
  - Primary probabilistic metric: measures both sharpness and calibration of the entire predictive distribution
  - Lower values indicate better probabilistic forecasts
- PICP (Prediction Interval Coverage Probability): 90% level
  - Percentage of actual values falling within the predicted interval
  - Well-calibrated if empirical coverage ≈ nominal (85-95% for 90% PI)

Probabilistic forecast metrics

Experimental Results

Average performance across 6 anchor dates (Mean ± Std)

Model	MAE (EUR/MWh)	RMSE (EUR/MWh)	CRPS (EUR/MWh)	PICP_{90} (%)
Naive	36.57 ± 26.21	45.38 ± 32.78	61.98 ± 91.41	54.7 ± 26.59
QRA	17.00 ± 14.70	22.87 ± 19.14	12.88 ± 11.34	86.1 ± 6.87
LQRA(BIC)	16.15 ± 13.03	21.67 ± 16.77	12.24 ± 10.01	85.7 ± 4.88
NGBoost	14.20 ± 9.55	19.12 ± 12.70	10.49 ± 7.22	75.2 ± 13.76



- » NGBoost: best accuracy (MAE: **14.20**, RMSE: **19.12**, CRPS: **10.49**) but under-calibrated intervals (**75.2%** vs. nominal 90%)
- » LQRA(BIC): optimal balance with high accuracy (only 14.3% behind NGBoost on CRPS) and well-calibrated uncertainty (**85.7%** coverage)
- » LQRA(BIC) vs. QRA: regularization provides marginal accuracy improvement (5.23% better CRPS) with better stability (**±4.88** vs. **±6.87** std)
- » High variability across market regimes (std up to **±91.41**) demonstrates robustness of multi-anchor testing approach across diverse conditions (stable 2018-19, crisis 2021-22, volatile 2024)

Experimental Results

Forecasting performance across anchor dates

	Anchor Date	Model	MAE (EUR/MWh)	RMSE (EUR/MWh)	CRPS (EUR/MWh)	PICP90% (%)
normal market	2018-09-01	Naive	14.04	17.55	10.22	57.92
		QRA	6.45	8.35	4.52	93.19
		LQRA(BIC)	6.71	8.50	4.74	88.19
		NGBoost	6.33	8.73	4.59	83.75
pre COVID	2019-12-01	Naive	9.57	13.06	7.22	62.22
		QRA	4.36	6.00	3.16	86.11
		LQRA(BIC)	4.26	5.96	3.15	83.61
		NGBoost	4.58	6.19	3.29	78.89
beginning energy crisis	2021-03-01	Naive	16.39	21.17	12.57	51.25
		QRA	8.72	12.26	6.27	88.89
		LQRA(BIC)	8.17	11.91	6.13	88.33
		NGBoost	9.67	13.12	6.79	93.89
peak crisis	2021-09-01	Naive	54.09	64.59	44.33	37.50
		QRA	40.93	52.67	31.51	76.11
		LQRA(BIC)	34.86	43.95	26.54	77.50
		NGBoost	22.37	29.02	16.96	53.61
high volatility	2022-06-01	Naive	70.50	93.37	244.38	99.03
		QRA	29.14	40.38	21.78	92.50
		LQRA(BIC)	30.04	41.26	22.96	91.53
		NGBoost	28.80	39.06	21.35	70.56
new normal	2024-03-01	Naive	54.79	62.53	53.14	20.14
		QRA	12.37	17.55	10.04	80.00
		LQRA(BIC)	12.85	18.40	9.88	85.14
		NGBoost	13.41	18.59	9.90	70.42

» Performance varies dramatically by market regime: best period **(2019-12)** shows CRPS of **3.15-3.29** EUR/MWh vs. crisis peak **(2021-09)** at **16.96-31.51** EUR/MWh, demonstrating 5-10× performance degradation during extreme volatility

» NGBoost achievers lowest CRPS in 3 of 6 periods, particularly excelling during crisis (2021-09: 36%-46% better than statistical methods)

» Critical calibration trade-off: NGBoost under-covers systematically (**53.61%-93.89%** vs. nominal 90%), while LQRA/QRA maintain more reliable uncertainty bounds (~76% - ~93% coverage) essential for risk management applications

» Naive method catastrophic failure during 2022-06 volatility (CRPS: **70.50** EUR/MWh, 2.5× worse than advanced methods) validates the necessity of probabilistic forecasting infrastructure for modern electricity markets

Take-home Messages

- NGBoost achieves the highest accuracy for electricity price forecasting with 10.49 EUR/MWh CRPS→ translating to additional revenue per MW trading position in the Dutch day-ahead market

- Crisis period performance validates NGBoost superiority: during 2021-09 peak volatility, NGBoost maintains 36% CRPS advantage over LQRA (16.96 vs. 26.54 EUR/MWh)→ leads to savings per MW when price prediction errors are most financially severe

- For profit-maximizing trading strategies, NGBoost's accurate point forecasts (14.20 MAE vs. 16.15 LQRA) outweigh calibration concerns: trading decisions depend primarily on expected prices rather than tail risk quantification, making 17% CRPS improvement worth the narrower prediction intervals

Take-home Messages

- NGBoost's under-coverage (75% vs. 90% nominal) translates to 50% more price "surprises" than expected→ inconvenient for risk-constrained battery operations requiring reliable uncertainty bounds

- Calibration instability during crisis: NGBoost coverage collapses to 53.6% (2021-09), meaning battery scheduling algorithms face violated risk constraints 46% of trading days, vs. LQRA's resilient 77.5%

- LQRA demonstrates 3x superior calibration stability (±4.9% vs. ±13.8%), maintaining consistent performance across all market regimes→ critical for automated systems that cannot dynamically recalibrate

- For battery scheduling applications: LQRA's reliable 85.7% coverage justifies accepting 17% worse CRPS, as under-coverage costs exceed accuracy gains through systematic constraint violations and increased operational risk

THANKS FOR WATCHING!

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